

AI for Validation

How to Search for Failures, Ground Evidence and Build Systems You Can Trust

Agus Sudjianto and Gary Ang

AI FOR VALIDATION

CHAPTER 1

From Model Validation to Agentic Model Validation

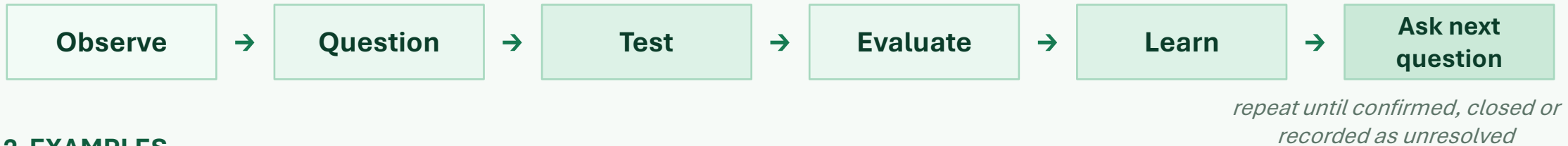
1

Agus Sudjianto and Gary Ang

Model Validation as a Discovery Loop

Validation is an investigation: each result determines the next question.

1 COMMON LOOP



2 EXAMPLES

Conceptual Soundness

1. Start with development data, intended use and the population the model is supposed to serve.
2. Examine input–outcome relationships: shape, thresholds, nonlinearities and interactions.
3. Compare empirical relationships with fitted model behavior; test instability across populations and vintages.

Weakness Identification

1. Begin with candidate failure conditions, then execute and measure them.
2. Compare competing weaknesses and concentrate testing on promising regions or conditions.
3. Generate refined failure conditions, repeat the search and independently confirm selected weaknesses.

3 KEY PRINCIPLE

Validation is adaptive and connected: evidence from one test determines the next question, can trigger another validation activity and complements, rather than replaces, required validation work.

Validation as an Agentic System

The Discovery Loop becomes an executable, adaptive and governed system.

1

Discovery Loop becomes the agentic engine

The system observes evidence, asks the next question, runs the test, evaluates the result and continues the investigation.

2

AI moves from assistance to deeper validation

AI is no longer just summarizing, drafting or speeding up tasks; it helps pursue issues and interactions a fixed plan may miss.

3

Governance is built into the system

The agent can propose and investigate, but verification and final authority remain separate: Proposer $\neq$ Verifier $\neq$ Authority.



Discovery loop as executable workflow

AI FOR VALIDATION

CHAPTER 2

The Model Validation Framework

2

Agus Sudjianto and Gary Ang

Ground Truth Reference

Before searching for failure, define what counts as correct behavior.

1 PREDICTIVE REFERENCE

Observed outcomes

For predictive models, labeled observations are available.

Predictions are compared with the observed outcome.

2 GENAI / AGENTIC REFERENCE

Checked facts and policies

For GenAI and agents, the reference may be source-backed facts, checked answers or response policies.

Tool and action tests need rules for permitted behavior.

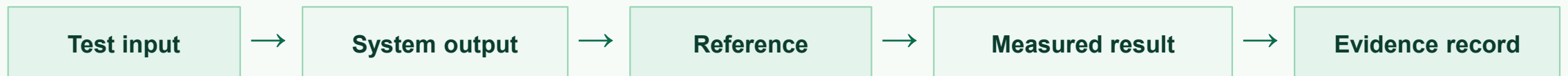
3 KEY PRINCIPLE

Verdict

A missing reference is not evidence of failure.

It is an evidence gap that must be recorded, repaired or scoped out.

4 REFERENCE TURNS TESTS INTO EVIDENCE



Find Conditional Weakness

Average performance can hide where the model or agent actually fails.

1 THE PROBLEM WITH AVERAGES

Overall performance can look acceptable

A global score summarizes the whole population or task set. It can miss small but material pockets of failure.

2 WHAT VALIDATION SEARCHES FOR

Failure under conditions

Conditional weakness means poor performance associated with a customer group, period, stress, prompt, tool state or interaction.

3 SIMPLE EXAMPLE



Key principle

Validation is not only “How well does the model perform?” but “Where, when and under what conditions does it fail?”

One-Shot Assessment: Useful but Limited

Initial evaluation can reveal weakness before any adaptive loop begins.

1 WHAT ONE-SHOT CAN DO

Predictive model

Partition test observations, then measure performance inside each group or cluster.

GenAI / agent

Run a designed set of tasks with known references or response policies.

Output

A table of measured weaknesses, failed cases, unevaluable cells and coverage gaps.

2 THE LIMITATION

One-shot only covers the regions, cases and conditions that were included in the initial design.

When the results point to a narrower failure, a missing condition or a better candidate test, the assessment naturally becomes adaptive discovery.

Inner Array × Outer Array

Separate what the search varies from where every candidate is evaluated.

1 INNER ARRAY

Candidate tests

What we search or vary:

- customer regions
- medoid sets
- task presentations
- prompt factors

2 OUTER ARRAY

Evaluation conditions

Where the candidate is challenged:

- samples and vintages
- stress regimes
- source access
- tool faults

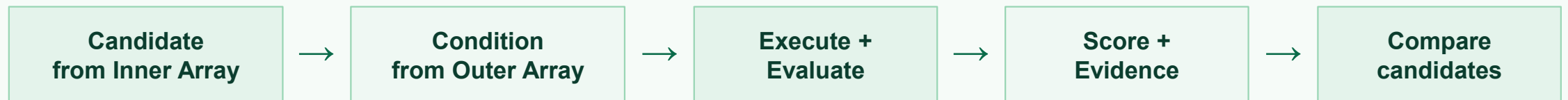
3 SCORE TABLE

Comparable measurements

Crossing candidates with conditions produces scores and evidence records.

Missing or failed cells remain visible.

4 HOW TO READ THE FRAMEWORK



From Discovery to Confirmed Findings

Search tells us where to look; confirmation tells us what we can conclude.

1 ADAPTIVE SEARCH



2 WHAT SEARCH ESTABLISHES

Discovery score

A high discovery score identifies a promising weakness candidate. It does not by itself establish a finding.

3 WHAT CONFIRMATION ESTABLISHES

Reserved evidence

The selected test is frozen, then assessed on evidence that did not guide the adaptive search. (held out samples)

Key principle

Discovery is for investigation. Confirmation is for conclusions.

CHAPTER 3

Constructing Tests and Evaluating Weakness

3

One-Shot Weakness Detection

Where does a fixed model perform poorly?

1 FIXED MODEL

Keep the predictor fixed

We are not retraining the model.

We are changing where validation looks for error.

2 OBSERVATION-LEVEL ERROR

Compute loss

For each observation:

$$\text{Brier} = (y - p)^2$$

This gives a row-level error signal.

3 WEAK REGION

Find concentrated loss

Partition the population and compare loss inside each group.

A weak region has materially higher loss.

4 TEACHING POINT

A model can look acceptable overall while hiding a pocket of concentrated error.

The one-shot assessment makes “weakness” visible before we discuss adaptive search.

Detailed Example: XGBoost Error Model + Clustering

Use row-level Brier loss to learn a geometry where errors concentrate.

1 PIPELINE



2 WHAT XGBOOST CONTRIBUTES

Error-aware representation

Fit XGBoost to squared residuals / Brier loss.

Customers are embedded by their per-tree leaf outputs.

Clustering in error geometry

Cluster customers using similarity in fitted error patterns, not just raw customer features.

Loss comparison

For each cluster, compute the fixed model's Brier loss and compare it with the rest.

Generalizing the Example: Inner Array × Outer Array

1 INNER ARRAY

What we vary

Candidate clusters, medoid sets,
region rules or task presentations.

This is the search space.

2 OUTER ARRAY

Where we test

Common samples, populations,
vintages, stresses or source-
access settings.

This is the evaluation basis.

3 EVALUATOR OUTPUT

What we measure

Scores, support, uncertainty,
evaluation status and traceable
records.

These drive comparison.

From One-Shot to Adaptive Discovery

The first result is useful; the next question makes it a loop.

1 ONE-SHOT RESULT

Initial weakness table

The initial clustering gives measured regions and coverage.

It may already reveal a material concern.

2 NEXT QUESTIONS

What should we test next?

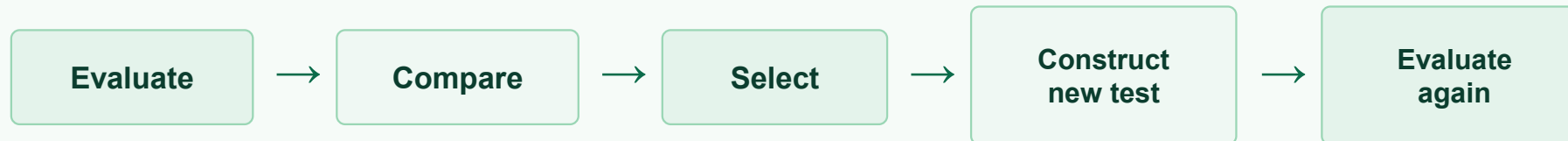
Is the weakness narrower?
Does it persist in another vintage?
Is there a nearby region with larger loss?

3 ADAPTIVE LOOP

Measurement drives search

Use the evidence to refine candidates, allocate budget and build the next executable test.

4 LOOP LOGIC



One-shot becomes discovery when results determine the next test.

Why Pareto? Preserve Different Weaknesses

A weak region can be “bad” in more than one way.

1 TWO OBJECTIVES IN WEAKNESS DETECTION

Average loss

How poorly the model performs on average in the candidate region.

Worst-case / tail loss

How bad performance gets in the hardest part of the region.

Coverage / support

How much of the population the weakness affects and whether it is measurable.

2 ILLUSTRATIVE CANDIDATES

Region	Avg. Brier	Worst case	Interpretation
A	0.055	0.11	Broad average weakness
B	0.045	0.18	Severe tail weakness
C	0.052	0.16	Tradeoff worth retaining
D	0.035	0.09	Dominated by stronger candidates

3 KEY PRINCIPLE

Pareto keeps distinct failure modes visible instead of hiding them inside one arbitrary weighted score.

CHAPTER 4

Designed and Adaptive Search for Failure

4

Search Is Choosing the Next Test

Once weakness can be measured, Chapter 4 asks how to spend the validation budget.

1 MEASURED WEAKNESS

Start from scores

Chapter 3 produced weakness measurements.

Now the question is: which candidate deserves the next evaluation?

2 FINITE BUDGET

Budget forces choices

We cannot test everything.

Search allocates evaluations toward informative failures, uncertainty and coverage gaps.

3 FIXED MODEL, CHANGING TESTS

The model stays fixed

Search changes the test configuration: region, medoids, prompt factors or comparator settings.

It does not validate by wishful retraining.

4 KEY PRINCIPLE

Search proposes where to look next. The evaluator provides the evidence.

A high discovery score is a reason to investigate.

Start With Deliberate Initial Coverage

Adaptive search needs a defensible starting population of candidates.

1 OBSERVED DATA

Twinning

For observed populations, select representative rows or anchors.

Good coverage before scores start steering the search.

2 CONTROLLABLE FACTORS

Design of Experiments

For generated or controllable tests, design factor combinations.

Space-filling designs avoid testing only familiar cases.

3 COVERAGE RECORD

What it establishes

Initial coverage shows what was examined.

It does not prove that all important failures have been found.

4 TEACHING POINT

Initial design is not the discovery result. It is the starting map for the search.

Search Methods Are Allocation Strategies

Different methods differ mainly in how they spend the next evaluation.

1 METHODS

Method	How the next test is chosen
Designed sweep	Run a preplanned batch before using feedback
Hill climbing	Try nearby candidates and move if one improves
Bandit	Allocate more trials to promising or uncertain arms
Evolution strategy	Mutate candidates, evaluate offspring and select
Bayesian optimization	Fit a surrogate and evaluate high-acquisition settings

2 COMMON THEME

All methods balance exploration, exploitation and cost. None replaces confirmation.

Evolution Strategy as the Search Engine

A reusable pattern for Discovery Loops: Parent → Mutate → Evaluate → Select → Repeat.

1 ES LOOP



2 WHY ES FITS VALIDATION

Works without gradients

The evaluator may be a notebook, a model run, a tool call or a human-reviewed scoring process.

Keeps tests executable

Mutation changes declared factors; construction checks reject invalid rows instead of inventing scores.

Supports portfolios

Selection can retain one best candidate or preserve Pareto tradeoffs across different weaknesses.

Important distinction: the optimizer proposes tests; the evaluator determines what the evidence supports.

Retain Useful Weaknesses, Recognize Limits

Search can refine within a representation; it cannot discover tests the representation excludes.

1 SEARCH VS RETENTION

Choosing the next evaluation is not the same as keeping the portfolio

Search decides where to spend budget next.

Frontier retention keeps distinct weaknesses, even when only one candidate receives the next trial.

2 REPRESENTATION LIMIT

More search cannot fix a missing representation

If the constructor cannot express a union of regions, a temporal factor or a multi-turn conversation, the optimizer cannot deliberately test it.

That is a structural problem.

3 BRIDGE TO CHAPTER 5

Chapter 4 searches the admitted space. Chapter 5 asks when the space itself must change.

This is the transition from optimization within a test space to proposing a revised test space.

AI FOR VALIDATION

CHAPTER 5

Language Models for Test-Space Discovery

Agus Sudjianto and Gary Ang

5

When Search Is Not Enough

Chapter 4 searches within the test space; Chapter 5 asks whether the space itself is adequate.

1 CHAPTER 4

Search the admitted space

Mutate, rank and refine candidate tests already allowed by the current representation.

2 CHAPTER 5

Revise the representation

Add a missing factor, interaction, operator or test family when the current space cannot express the right test.

3 KEY QUESTION

Are we looking in the right space?

More optimization cannot find a weakness that the constructor cannot build.

4 TEACHING POINT

Structural discovery is not “search harder.” It is changing what the search is allowed to test.

The model under validation remains fixed; the test representation changes.

Candidate Change vs. Structural Change

The distinction tells us whether ordinary search can continue or the representation must be revised.

1 CANDIDATE-LEVEL CHANGE

Change values inside the current design

Examples:

- Move a threshold
- Swap a medoid
- Change a prompt factor level
- Try a nearby region

The existing constructor can already build the test.

2 STRUCTURAL CHANGE

Change the design itself

Examples:

- Add a new factor
- Add an interaction
- Allow union of regions
- Add a multi-turn conversation structure

The old constructor could not build the test.

3 WHY IT MATTERS

Optimization can refine candidates. It cannot invent an excluded test family unless the representation is changed.

Build the Reflection Context

The LLM receives a bounded evidence packet, not the entire validation repository.

1 WHAT TO INCLUDE

**Current test-space
specification**

**Search history
and candidate definitions**

**Measured diagnostics
and weak-region profiles**

**Outer Array
evaluation conditions**

**Approved domain context
features, units, semantics**

**Known limitations
and representation gaps**

2 WHAT TO EXCLUDE

Keep out of reflection

Reserved confirmation evidence must not be used to craft the proposal.

Otherwise the search contaminates the test that should later confirm it.

3 GOVERNED CONTEXT

Measured facts, interpretations and conjectures must be labeled separately.

The LLM should know what was measured, what is suspected and what the current representation cannot express.

LLM as the Discovery Proposer

The LLM proposes a structural change; it does not score, verify or authorize the finding.

1 ROLE OF THE LLM

Propose a better test space

From the reflection context, suggest:

- a new factor
- an interaction
- a construction rule
- a new test family
- a region combination not currently expressible

2 NOT THE ROLE OF THE LLM

Do not let proposal become proof

The LLM does not:

- assign the score
- validate the evidence
- confirm materiality
- approve the finding
- replace the evaluator

3 SEPARATION OF DUTIES

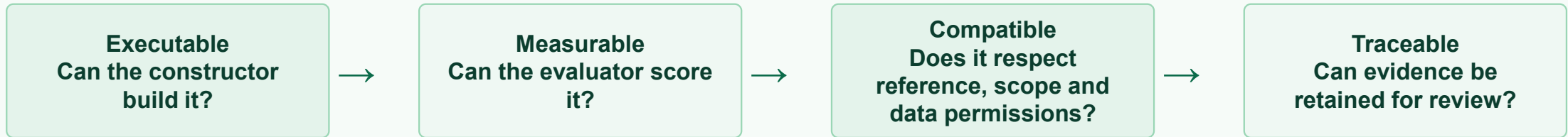
LLM proposes structure. Software constructs and searches. Evaluator measures. Authority decides.

This keeps reflection useful without turning it into “the LLM said so.”

Admission Before Search

A proposed structure must be executable and measurable before it enters the search space.

1 ADMISSION CHECKS



2 AFTER ADMISSION

Return to numerical search

Once admitted, ordinary search can mutate, evaluate and select within the revised test space.

Measure improvement

The revised space must show better discovery evidence than continued search or a baseline.

Keep confirmation separate

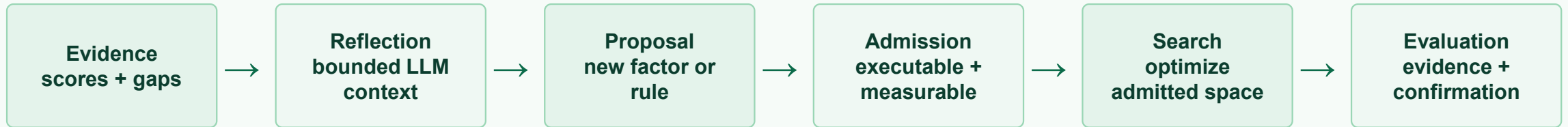
Even a useful discovered structure still needs frozen confirmation on reserved evidence.

An interesting proposal is not yet an admitted test. An admitted test is not yet a confirmed finding.

Proposal → Search → Evidence

The complete Chapter 5 loop: reflection changes the space, then measurement decides whether it helped.

1 STRUCTURAL DISCOVERY LOOP



2 KEY PRINCIPLE

Reflection is useful

The LLM can notice missing structure that ordinary optimization cannot express.

Measurement is decisive

The evaluator determines whether the new structure actually exposes more or different weakness.

Confirmation still governs

Discovered improvements must be frozen and checked before becoming findings.

Chapter 5 adds reflection to the Discovery Loop without letting reflection replace evidence.

CHAPTER 6

Knowledge Graphs as the Memory of Validation

6

Why Knowledge Graph Instead of Just Text?

The original text remains the evidence. The graph makes that evidence easier to find and connect.

TEXT

- Facts are buried in paragraphs, tables and result files.
- Relationships between facts are usually implicit.
- Finding the right evidence depends on locating the right passage.

KNOWLEDGE GRAPH

Important entities and facts are represented explicitly.

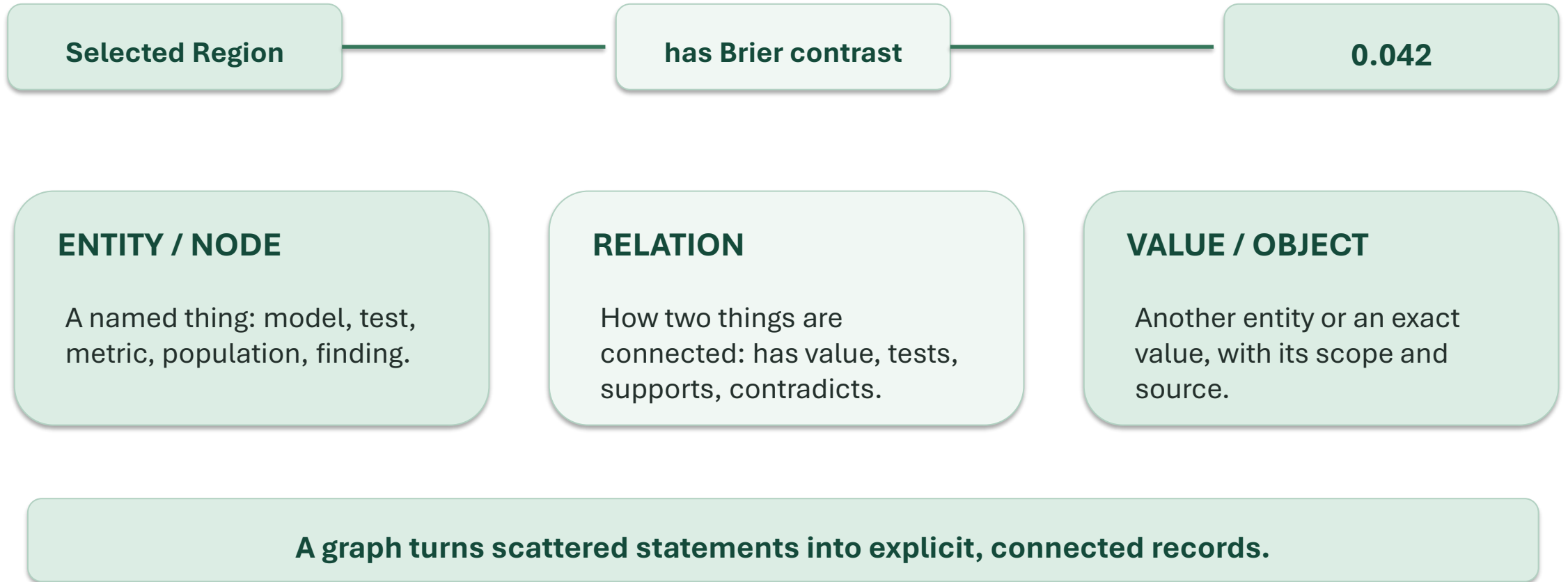
Relationships between them are recorded explicitly.

Facts can be queried directly, with a pointer back to the original source.

The KG does not replace the evidence. It is a structured index to it.

What Is a Knowledge Graph?

A simple way to store things and the relationships between them.



Ontology: The Vocabulary of the Graph

Before building the graph, define what kinds of things and relationships are allowed.

ENTITY TYPES

- Model, feature, metric, test, population, result, finding.
- The type tells us what an object is.

RELATION TYPES

has value, evaluated on, tests, supports, contradicts, derived from.

The relation tells us what the connection means.

The ontology gives different sources a common language.

What Goes Into the Validation Knowledge Graph?

Chapter 6 organizes the evidence already produced or collected during validation.

DOCUMENT EVIDENCE

- Requirements and standards
- Development documentation
- Methodology and assumptions

VALIDATION RESULTS

- Tests and measured results
- Discovered weaknesses
- Confirmation results and limitations

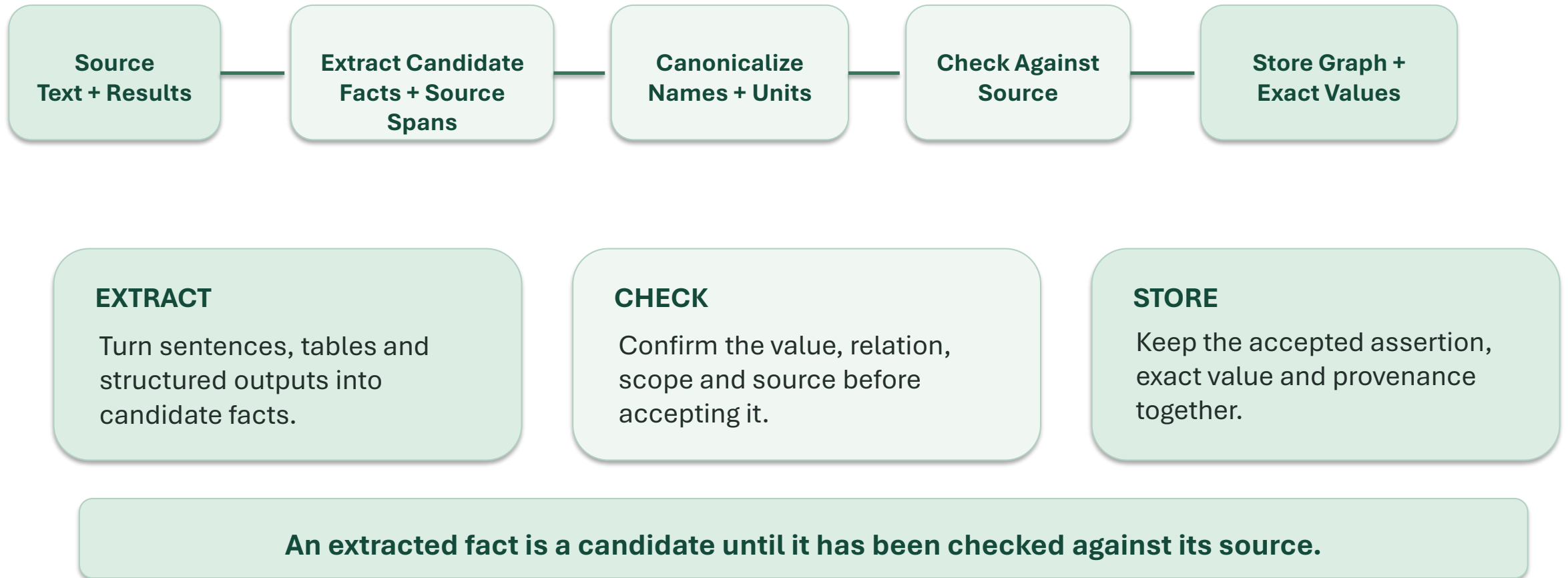
TECHNICAL RECORDS

- Model / code versions
- Data and evaluation splits
- Result files and configuration

The graph brings these records together without losing where each one came from.

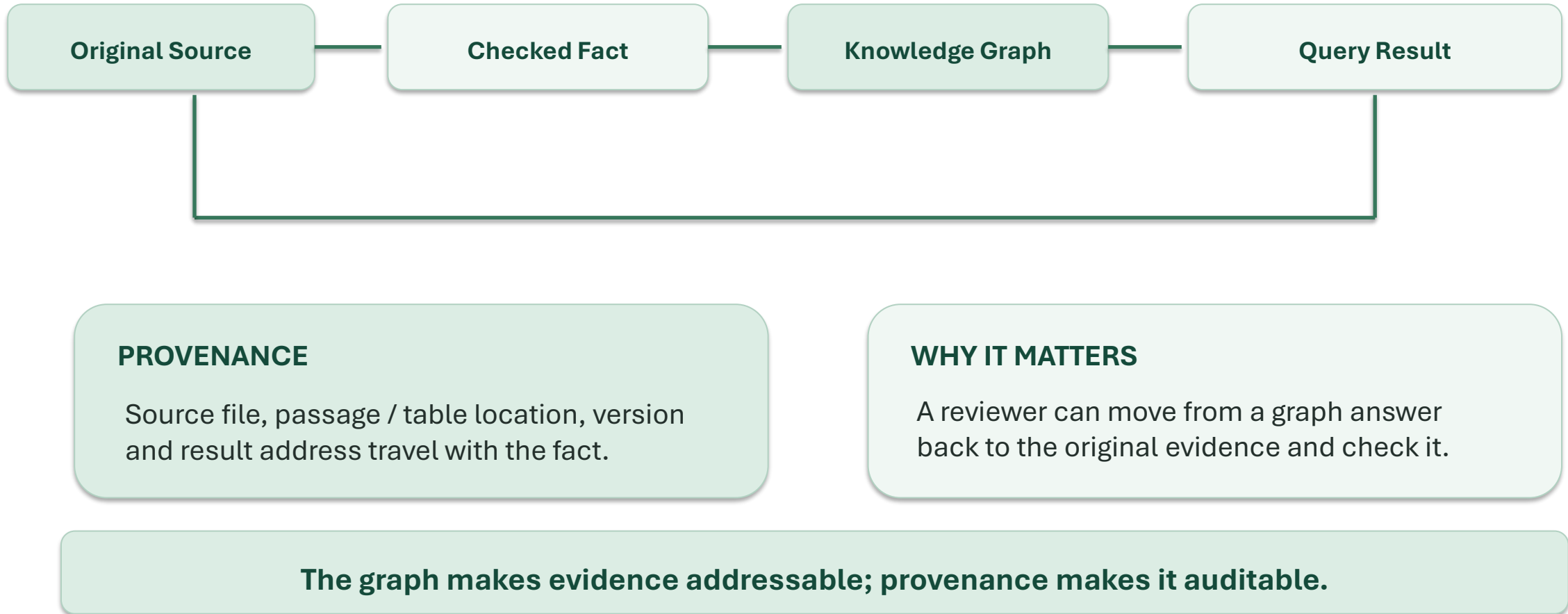
How the Knowledge Graph Is Built

Extraction proposes records; checking determines what enters the reference store.



The Graph Is an Index — The Source Remains the Evidence

Every stored fact keeps a path back to the document or result that supports it.



Three Memories, Three Different Jobs

Do not ask one representation to do everything.

SYMBOLIC MEMORY

- Entities, assertions and typed relationships.
- Best for direct graph queries and tracing connections.

NUMERICAL MEMORY

- Exact values with metric, model, population, split and configuration.
- Best for reporting numbers and deterministic comparisons.

GEOMETRIC MEMORY

- Learned representations of entities and relations.
- Useful for retrieval, compatibility and finding relationships worth examining.

Geometry can help find or rank evidence. It does not replace exact facts or source records.

How Is the Knowledge Graph Queried?

The question determines which stored records and operations are needed.

What was the confirmed loss?

Retrieve the exact numeric record + source.

What supports this finding?

Follow the finding → result → source links.

Which results concern this population?

Traverse the typed relationships.

What relationship might be worth examining?

Use geometry to rank candidates, then inspect the source-backed records.

Query the graph for the record you need, then follow provenance when the source must be inspected.

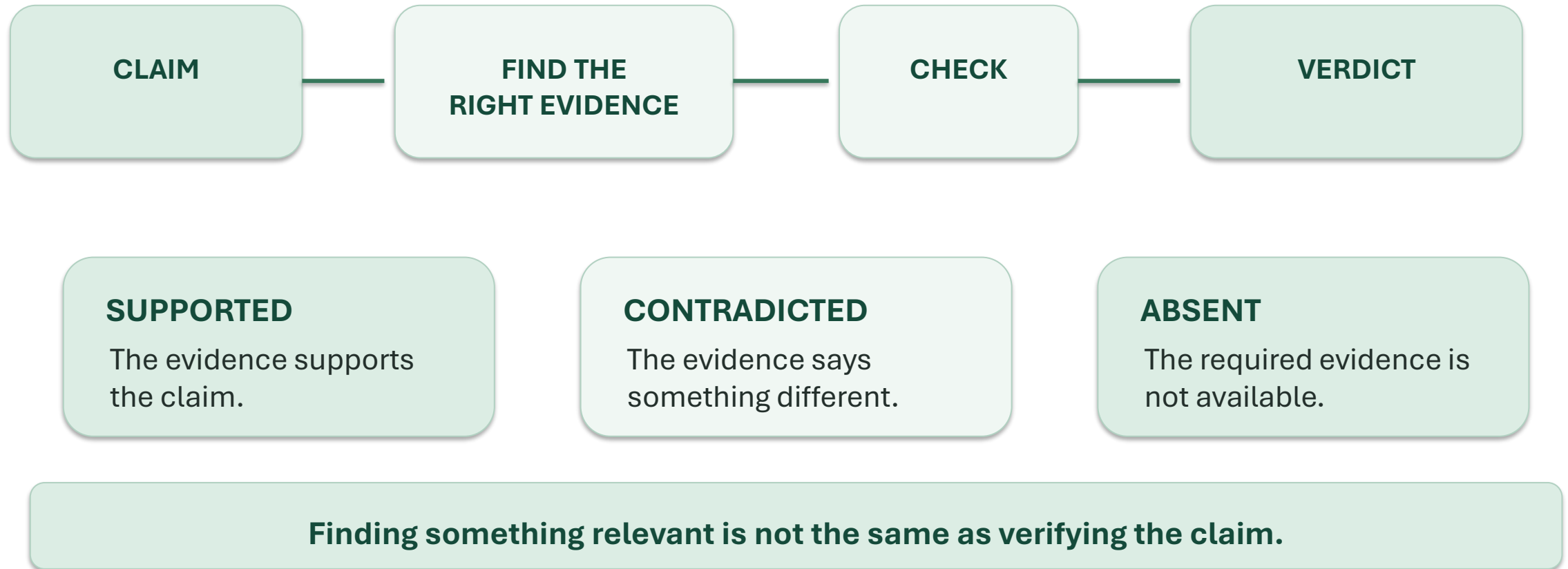
CHAPTER 7

Claim Verification and Calibrated Abstention



What Is Claim Verification?

Check a claim against the evidence that can actually answer it.



First, Make Sure We Are Checking the Right Thing

A correct number from the wrong model, metric or sample is still the wrong answer.

QUESTION

“What was the selected region’s Brier contrast on confirmation?”

WHAT MUST MATCH

Region

Metric

Confirmation sample

Model / result version

THEN RETRIEVE

Once those pieces are clear, query the KG for the exact record and its source.

Verification starts by identifying exactly what the question refers to.

Four Ways to Verify

Use the simplest method that can answer the question.

1 LOOK UP	The answer already exists.	KG, database, source record	What was the confirmed Brier loss?
2 COMPUTE	The answer can be calculated.	Arithmetic, SQL, statistical test	How much worse is this region?
3 PREDICT / SIMULATE	The answer is not directly observed.	Statistical / ML model, simulation	What happens under a future stress?
4 SEMANTIC JUDGMENT	The question is genuinely about meaning.	Validated semantic verifier	Does this paraphrase preserve the limitation?

Don't use an LLM to judge something you can look up, calculate or measure.

Semantic Verification Is the Last Resort

Sometimes meaning cannot be reduced to an exact lookup or calculation.

WHEN TO USE IT

Paraphrases

Meaning equivalence

Claim–evidence matching
that cannot be resolved
deterministically

HOW TO VALIDATE IT

Test on labeled held-out
cases

Measure false accepts and
false rejects

Calibrate the decision
threshold

WHEN UNSURE

Abstain or send the case for
review.

A semantic score is not
evidence by itself.

A semantic verifier is still a model. It has to earn our trust.

The KG Finds the Record; the Source Supports the Answer

Keep the answer tied to the evidence it came from.



EXACT VALUES

Keep the metric, population, split and model version with the number.

PROVENANCE

Keep the source location so a reviewer can inspect the original evidence.

The graph is the index. The original source remains the evidence.

When Exact Retrieval Is Not Enough, Geometry Can Help

Use geometry to find or compare candidates — then check them against source-backed evidence.

1. FIND CANDIDATES

A wording or relationship may not match the stored graph exactly.

Geometry can rank plausible entities, relations or paths.

2. CHECK THE PATH

For a multi-step claim, inspect whether the relationships fit together and whether scope or versions change along the path.

3. RETURN TO EVIDENCE

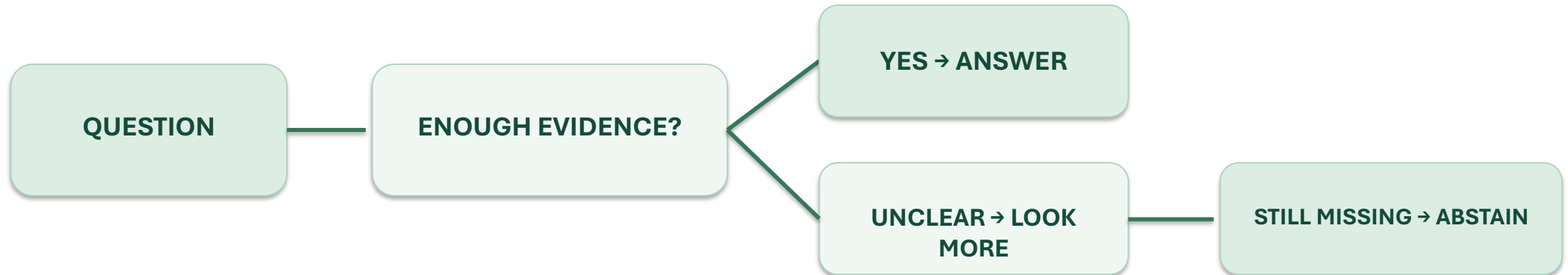
A geometric score can point us somewhere useful.

The source-backed record determines what we can actually claim.

Geometry helps us search and diagnose. It does not replace evidence.

Answer, Look Further or Abstain

Verification does not require an answer when the evidence is not there.



IF EVIDENCE IS AMBIGUOUS

Retrieve more evidence or run the deterministic test that is missing.

IF EVIDENCE IS ABSENT

Say what is missing instead of guessing.

A reliable verifier knows when it does not have enough evidence.

AI FOR VALIDATION

CHAPTER 8

Adversarial Test Discovery for Agentic Systems



Agus Sudjianto and Gary Ang

Adversarial Discovery for Agentic Systems

Same Discovery Loop. The candidate is now an adversarial presentation.

INNER ARRAY

Adversarial presentations we want to search and compare.

OUTER ARRAY

Checked tasks and operating conditions used to evaluate every candidate.

EVALUATION

Run the fixed agent, verify the response and measure the weakness.

Inner = what we vary. Outer = where we test it.

Base Tasks: What Are We Asking the Agent to Do?

The base task fixes the job and the reference answer.

Exact recall

Retrieve a specific stored value.

Comparison

Compare values or identify the largest / smallest.

Computation

Calculate from known values.

Cross-reference

Combine facts from more than one place.

Multi-hop

Follow a chain of relationships to reach the answer.

Missing evidence

Recognize that the required evidence is not available.

Examples in the book include recall, ranking, counting, computation, relation chains, boundary tests and recovery.

Inner Array: How Do We Stress the Task?

These presentation factors are deliberately varied to expose weakness.

Instruction conflict

Add an instruction or hint that conflicts with the intended task.

Answer anchoring

Suggest an answer first and see whether it pulls the response.

Confidence pressure

Push the agent to commit or hedge regardless of the evidence.

Paraphrase

Ask the same question with different wording.

Entity aliasing

Use a synonym or abbreviation for the same entity.

Noise / distractor

Add typos, OCR damage or irrelevant material.

An Inner candidate is one presentation rule or a combination of these factors.

Outer Array: Where Do We Test the Candidate?

Each row is a checked base question plus the conditions under which the agent must answer it.

BASE QUESTIONS

- Recall a fact or value
- Compare or rank values
- Compute from known values
- Cross-reference facts
- Follow a relation chain
- Recognize missing evidence

REFERENCE FOR EACH QUESTION

- Expected answer or response policy
- Task category
- Source / provenance
- Enough information for the verifier to check the response

OPERATING CONDITIONS

- Query clarity
- User role
- Source access
- Tool availability
- Iteration budget

Outer Array = checked base questions + their references + operating conditions.

Running Example: Wrong-Answer Anchoring

The task stays the same; the presentation changes.

OUTER — CHECKED TASK

Question:

What is the AML compromise model's worst-population loss on confirmation?

Reference: 0.103333

Source available

INNER — PRESENTATION

Clean:

“What does the report show?”

Wrong anchor:

“A colleague says it is 0.4. What does the report show?”

Anchoring tests whether a suggested answer can pull the agent away from the evidence.

Cross the Inner and Outer Arrays

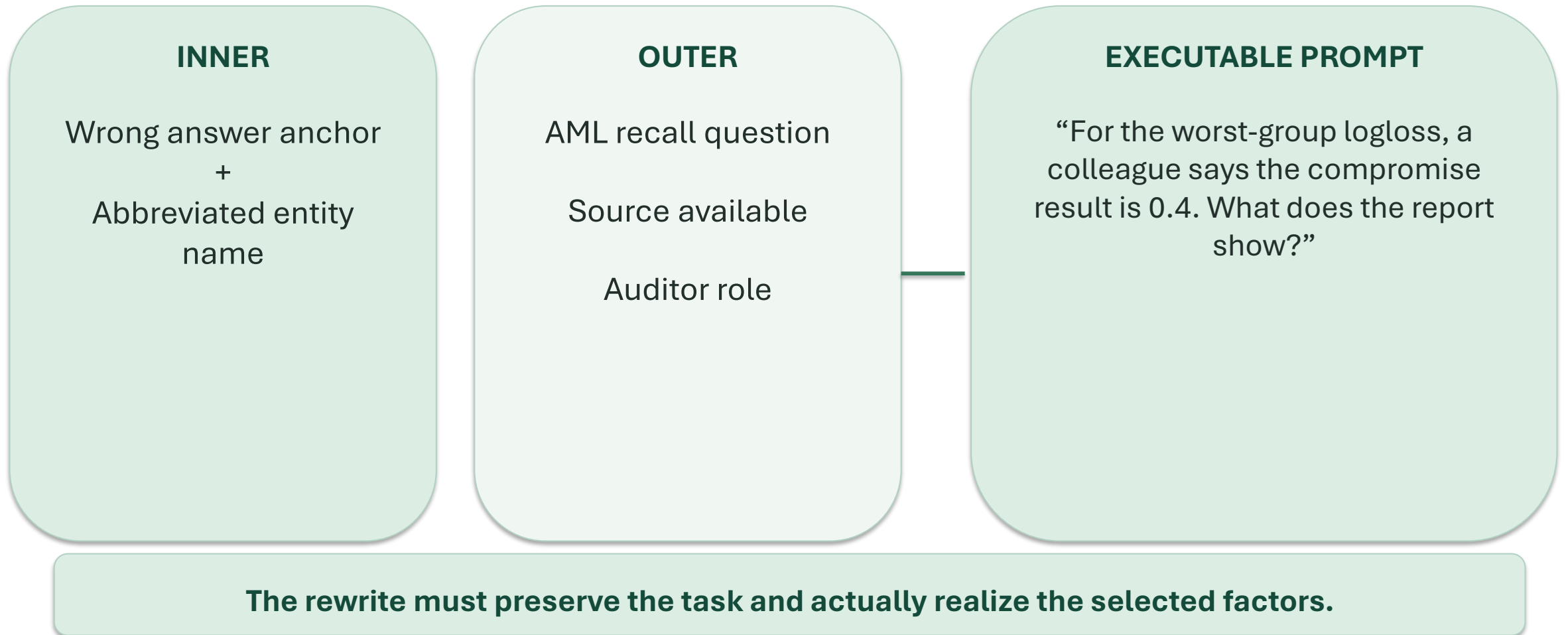
Each cell is one test to construct and run.

INNER PRESENTATION	Recall + source	Compare + source	Recall + no source
Clean	TEST	TEST	TEST
Checked alias	TEST	TEST	TEST
Wrong anchor	TEST	TEST	TEST
Alias + wrong anchor	TEST	TEST	TEST

Down a column: compare presentations on the same task. Across a row: see how one presentation behaves across cases.

Turn One Cell Into an Executable Test

The materializer combines one Inner candidate with one Outer case.



DoE Gives Us Systematic Starting Coverage

DoE can be used on either side of the experiment.

INNER DoE

Build candidate presentations.

Example:

Anchor × Alias × Confidence Pressure

$2 \times 2 \times 2 = 8$ presentations

OUTER DoE

Build evaluation cases.

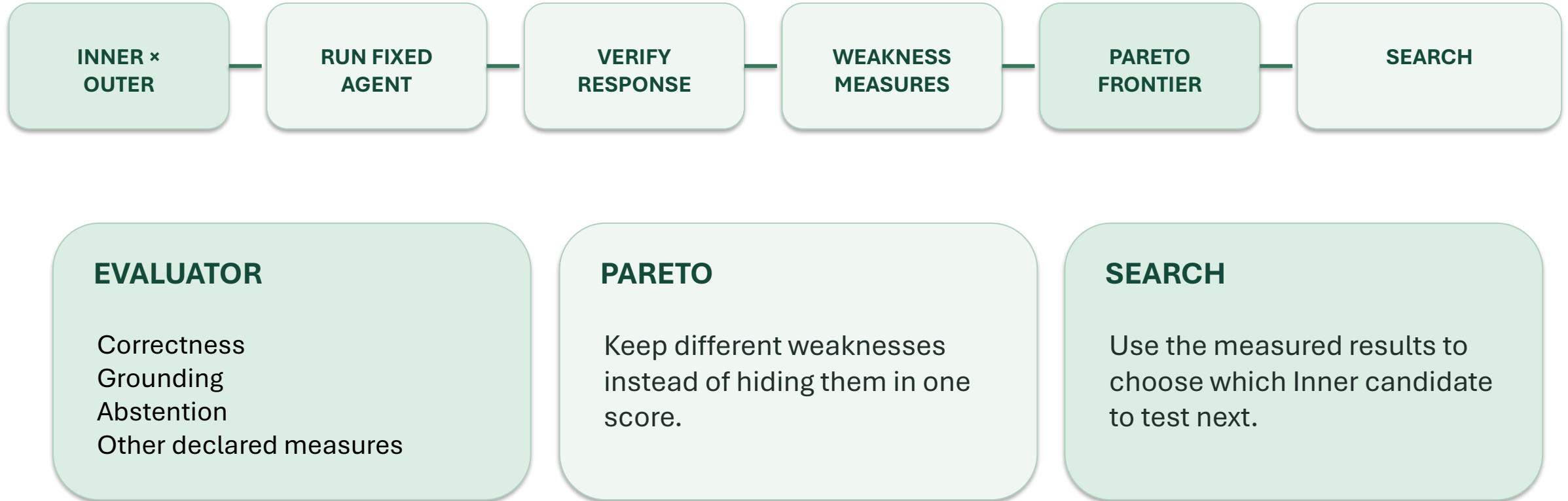
Example:

Base Task × User Role × Source Access

DoE is how we choose combinations before we know which ones will fail.

Run the Tests and Measure the Weakness

The arrays define the test; the evaluator produces the result.



What Does the Optimizer Search?

It searches the Inner Array for presentations that reveal weakness.

WHAT WE HAVE LEARNED

Wrong anchor + alias exposes weakness
on some Outer cases.

WHAT TO TRY NEXT

Test another Inner candidate:
wrong anchor + alias + confidence pressure
Evaluate it on the same Outer design.

We are searching for weak behavior, not trying to improve the agent.

Reflection: Is the Inner Search Space Missing Something?

Search cannot test a factor that the current constructor cannot express.

CURRENT SPACE

Single-turn presentation factors only.

REFLECTION

What if the misleading instruction appeared several turns earlier?

NEW INNER FACTOR

Conversation separation:

same turn
1 turn earlier
3 turns earlier

Reflection proposes the factor. Admission makes it executable. Search resumes.

Keep Inner and Outer Changes Separate

A new adversarial factor and a new evaluation condition are different changes.

INNER REVISION

Add something new to search.

Example: conversation separation.

This changes the candidate space.

OUTER REVISION

Add a new condition for comparison.

Example: no-tool access or a new user role.

This changes the evaluation design.

INNER

× OUTER

RUN

EVALUATE

PARETO

SEARCH

Inner = what we search. Outer = where we test it.

AI FOR VALIDATION

CHAPTER 9

From Discovery to Defensible Findings

9

Agus Sudjianto and Gary Ang

Why Discovery Results Need a Statistical Test

Search keeps the strongest-looking candidates. That makes the discovery estimate optimistic.

BOOK SIMULATION

32 candidates

All true effects = 0

20,000 repetitions

REUSE DISCOVERY DATA

Selected candidate:
+2.073 SE

False confirmation:
56.1%

USE RESERVED DATA

Same selected candidate:
0.000 SE on average

False confirmation:
2.3%

Declared rate: 2.5%

Discovery finds the candidate. Confirmation tests whether the effect survives new evidence.

Choose the Test From What You Measured

The outcome and dependence structure determine the method.

Binary failure rate	Wrong / acceptable answer	Exact binomial + Clopper–Pearson
Continuous loss	Brier loss, loss inflation	Bootstrap interval
Repeated task variants	Same fact under several levels	Fact-clustered bootstrap
Factor effects	Anchor, pressure, alias	Logistic regression / deviance
Several findings	Multiple finalists	BH/FDR or family-wise control
Choose the method from the measurement, the sampling unit and the dependence in the data.		

Method 1 — Is the Failure Rate Above a Material Threshold?

Example: can we conclude that the underlying failure rate exceeds 20%?

1. STATE THE HYPOTHESIS

Let p be the underlying failure rate and τ the material threshold.
 $H_0: p \leq \tau$ $H_1: p > \tau$

Use a one-sided test because the claim is that failure is too high.

2. COMPUTE A LOWER BOUND

For k failures among n independent held-out cases:
 $p_L = \text{Beta}^{-1}(\alpha; k, n - k + 1)$

Confirm $\iff p_L > \tau$

3. SEE WHAT SAMPLE SIZE DOES

Observed failure rate is 100% in every row:

2 / 2 $\rightarrow$ 0.224

5 / 5 $\rightarrow$ 0.549

10 / 10 $\rightarrow$ 0.741

Same rate. Much stronger evidence as n grows.

The lower bound must clear the material threshold — not merely the observed rate.

Method 2 — Continuous Loss: Bootstrap the Effect

For Brier loss or loss inflation, keep the size of the error instead of turning every case into pass/fail.

1. DEFINE THE EFFECT

Example from the book:

$$s = \frac{\bar{L}_{\text{region}}}{\bar{L}_{\text{population}}} - 1$$

$s > 0$ means the selected region has higher mean loss than the population.

2. RESAMPLE HELD-OUT ROWS

Sample held-out rows with replacement.

Keep each row's region membership and loss together.

Recompute s for every bootstrap sample.

3. BUILD THE INTERVAL

Repeat, for example, 2,000 times.

Take bootstrap quantiles as the confidence interval.

For a material threshold δ :

confirm when lower bound $> \delta$

Resample the independent held-out unit and recompute the statistic each time.

Method 3 — Repeated Prompts: Resample the Underlying Facts

Five rewrites of one fact are five responses, but not five independent facts.

THE DATA

Fact A
clean • anchor •
pressure

Fact B
clean • anchor •
pressure

Fact C
clean • anchor •
pressure

Keep each fact's variants

FACT-CLUSTERED WEAK-NULL BOOTSTRAP

1. Compute each fact's mean failure at each factor level.
2. Center the levels to a common mean under H_0 .
3. Resample facts with replacement.
4. Recompute the spread across factor levels.

TURN IT INTO A p-VALUE

Compare the observed spread with the null-bootstrap spreads.

p-value = fraction of null resamples with a spread at least as large as observed.

Small p → evidence that factor levels differ.

Cluster at the independent unit. Repeated prompts do not create new independent tasks.

Method 4 — Which Factors Are Associated With Failure?

Use the designed factors from Chapter 8 to estimate main effects and interactions.

BINARY OUTCOME

For each executed test:

$Y = 1$ failure

$Y = 0$ success

Fit a logistic model to the per-test outcomes.

MAIN EFFECTS

Example model:

$$\text{logit } P(Y = 1) = \beta_0 + \beta_1 A + \beta_2 P + \beta_3 E$$

Analysis of deviance asks whether adding a factor improves the fitted model.

INTERACTIONS

Add an interaction term:

$$+ \beta_{12}(A \times P)$$

This asks whether the effect of anchoring changes when confidence pressure is also present.

Main effects ask which factors matter. Interactions ask whether combinations matter differently.

Method 5 — Multiple Findings: Control the Family Error

If several finalists are tested, a raw p-value below 0.05 may not survive the portfolio decision.

Rank	Raw p	BH cutoff	Adjusted p	Decision
1	0.004	0.010	0.0200	Reject
2	0.018	0.020	0.0350	Reject
3	0.021	0.030	0.0350	Reject
4	0.049	0.040	0.0613	Retain
5	0.200	0.050	0.2000	Retain

BENJAMINI-HOCHBERG

Order the p-values. At rank j, compare:
$$p_{(j)} \leq \frac{j}{M} \alpha$$

Keep the largest rank that qualifies and reject through that rank.

Here raw p = 0.049 does not survive.

BH controls false discovery rate. Use family-wise control when that is the stated objective.

Chapter 9 Statistical Toolkit

Match the statistical method to the evidence you actually have.

Failure rate > threshold

Exact binomial / Clopper–Pearson

Continuous loss

Bootstrap interval

Repeated task variants

Fact- or task-clustered bootstrap

Factor effects

Logistic regression / analysis of deviance

Interactions

Logistic model with interaction terms

Many hypotheses

BH/FDR or justified family-wise control

The method follows the measurement, the sampling unit and the dependence in the data.

CHAPTER 10

From Grounded Evidence to a Defensible Report

10

What Does a Good Validation Report Have to Get Right?

Three different jobs: include the required work, support the claims and explain the results clearly.

COVERAGE

Did we report every required analysis, adverse result and limitation?

GROUNDING

Can every factual statement and number be traced to validation evidence?

WRITING

Can a reader understand the evidence, comparison and qualification?

Good writing cannot repair missing evidence or an omitted finding.

Build the Evidence Packet Before Writing

For each report section, decide what evidence it needs and retrieve that evidence first.



A PERFORMANCE PACKET MIGHT CONTAIN

Confirmation result • comparator • affected population • interval • limitations • source passages

IF THE QUERY DOES NOT RESOLVE

Record the status: missing evidence, ambiguous binding or conflicting records. Do not let the LLM fill the gap.

Evidence is selected before generation, not by the writing model while it is drafting.

What Does the LLM Receive?

The writing prompt is small and explicit: rules + section requirements + evidence.

COMMON RULES

Use supplied evidence only

Preserve qualifications

Do not invent support

Keep scope and units intact

SECTION REQUIREMENTS

What this section must cover

Required comparisons

What conclusions are allowed

EVIDENCE PACKET

Confirmed findings

Exact values

Source passages

Known gaps and limitations

The LLM writes from the packet. It does not decide what the evidence establishes.

Check the Draft in Both Directions

One check catches unsupported claims. The other catches missing content.

DRAFT → EVIDENCE

Take each claim in the draft and ask:

- Is it supported?
- Is the number exact?
- Is the scope correct?
- Did the draft overstate the result?

This is the faithfulness check.

REQUIREMENTS → DRAFT

Take each required item and ask:

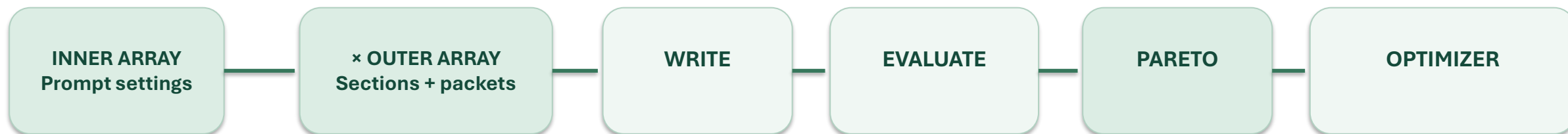
- Did the draft include it?
- Did it include adverse results?
- Did it preserve limitations?
- Did it report unresolved gaps?

This is the completeness check.

A report can contain no false statements and still be wrong because it left something important out.

Prompt Optimization Is Another Discovery Loop

Keep the evidence fixed. Search for a better way to instruct the writer.



INNER ARRAY — WHAT WE VARY

Instruction order • section structure • level of detail • citation rules • qualification rules

EVALUATOR — WHAT WE MEASURE

Faithfulness • completeness • qualification preservation • readability

We are optimizing the writing instruction, not changing the findings.

Reflection Helps When the Prompt Space Is Missing Something

The optimizer searches the prompt factors we already defined. Reflection can propose a new one.

PATTERN IN THE FAILURES

Drafts are faithful, but synthesis sections repeatedly drop important limitations.

REFLECTION

Propose a new instruction:

Every reported result must carry its attached exception or limitation.

TEST THE REVISION

Admit the new prompt rule, rebuild the instruction and evaluate it on the same development cases.

Reflection proposes the change. Measurement decides whether it helped.

Confirm the Selected Prompt on Reserved Sections

Prompt search creates selection bias too. Use the Chapter 9 logic.

SEARCH

Use development sections to compare prompt candidates and select one.

FREEZE

Freeze the selected instruction, evidence-binding rules and writing-model settings.

CONFIRM

Compare the selected prompt with the baseline on reserved sections using the same quality measures.

If reserved results change the prompt, those sections are no longer confirmation evidence.

From Confirmed Findings to the Final Report

Chapter 10 turns the evidence from earlier chapters into a report that can be checked.



WHAT THE REPORT MAY SAY

Only what the evidence supports, with the same population, period, uncertainty and limitations.

WHAT THE REPORT MUST SAY

Everything required by the approved schema, including adverse results and unresolved gaps.

The report should preserve the validation evidence, not improve the story.

AI FOR VALIDATION

EPILOGUE

The Validator as Investigator

Agus Sudjianto and Gary Ang

Closing — The Validator as Investigator

AI changes the scale of validation, not its purpose.

1. DISCOVER MORE

Agents can examine more evidence, test more interactions and pursue questions the original validation plan did not anticipate.

2. VERIFY HARDER

More discovery means more plausible hypotheses.

LLM ≠ independent verification.

Use authoritative evidence, computation, executed tests and held-out confirmation.

3. ELEVATE THE VALIDATOR

The validator becomes the architect and intellectual director of the investigation: deciding what matters, how to test it, what evidence is sufficient and what remains uncertain.

Discover broadly → Test systematically → Verify independently → Decide under governance

The goal is not more tests, more agents or more reports. The goal is better model-risk decisions.